Chapter 11. Smoothing and Density Estimation

Histograms, kernel density estimation, bandwidth selection, multivariate KDE, confidence bands

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11.1 From histograms to kernels

The empirical distribution function estimates F well but has no derivative, so it gives no direct estimate of the density f. A histogram does, but it depends on both the bin width and the bin origin. Averaging histograms over shifted origins gives a smooth estimate: the kernel estimator.

Definition (Kernel density estimator)

For a kernel K and bandwidth $h > 0$, $\hat{f}(x) = \dfrac{1}{nh}\sum_{i=1}^{n} K\Big(\dfrac{x - X_i}{h}\Big)$.

Standard kernels satisfy $K \geq 0$, $\int K = 1$, $K(-u) = K(u)$, and $\sigma_K^2 = \int u^2 K < \infty$. Examples are the Gaussian ϕ, the Epanechnikov kernel $\frac{3}{4}(1-u^2)_+$ and the uniform kernel $\frac{1}{2}I(|u| \leq 1)$. Then $\hat{f} \geq 0$ and $\int \hat{f} = 1$ (substitute $u = (x - X_i)/h$), so $\hat{f}$ is a density. The choice of h matters far more than the choice of K.

Example 11.1

Data $\{0, 1, 3\}$, Gaussian kernel, $h = 1$: $\hat{f}(1) = \frac{1}{3}[\phi(1)+\phi(0)+\phi(2)] = \frac{1}{3}(0.242+0.399+0.054) = 0.232$. The point 3, two bandwidths away, contributes little.

11.2 Bias, variance and bandwidth

Motivation. A kernel density estimate is controlled by its bandwidth: small bandwidths give noisy estimates, large ones oversmooth. Computing bias and variance shows how the optimal bandwidth shrinks with n and gives the $n^{-4/5}$ rate.

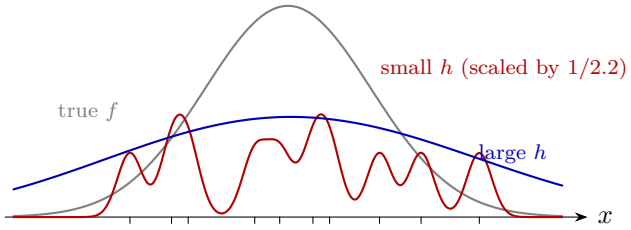


Figure. *Kernel density estimates from the same ten points (ticks). A small bandwidth gives a spiky estimate with large variance (its vertical scale is shrunk here to fit); a large bandwidth flattens the peak, a bias. The optimal $h \propto n^{-1/5}$ balances the two.*

Theorem 11.1

Let f be twice continuously differentiable near x with $f(x) > 0$. As $n \to \infty$, $h \to 0$ and $nh \to \infty$,

$$\operatorname{bias}\hat{f}(x) = \frac{\sigma_K^2 f''(x)}{2}h^2 + o(h^2), \qquad \operatorname{Var}\hat{f}(x) = \frac{f(x)R(K)}{nh} + o\Big(\frac{1}{nh}\Big), \qquad R(K) = \int K^2.$$

Proof [Supplement]. $E\hat{f}(x) = \int \frac{1}{h}K(\frac{x-t}{h})f(t)\,dt = \int K(u)f(x-hu)\,du$. Expand $f(x-hu) = f(x) - huf'(x) + \frac{h^2u^2}{2}f''(x) + o(h^2u^2)$, use $\int uK = 0$ and $\int u^2K = \sigma_K^2$, and justify the remainder by dominated convergence. For the variance, $\operatorname{Var}\hat{f}(x) = \frac{1}{n}\operatorname{Var}\big(\frac{1}{h}K(\frac{x-X}{h})\big)$. Here $E\frac{1}{h^2}K^2(\frac{x-X}{h}) = \frac{1}{h}\int K^2(u)f(x-hu)\,du = \frac{f(x)R(K)}{h} + O(1)$, while the squared mean is $O(1)$. ∎

This is the **bias–variance trade-off**: a small h gives low bias and high variance (a wiggly estimate), and a large h the reverse. Integrating,

$$\mathrm{MISE}(\hat f) = E\int (\hat f - f)^2 \approx \frac{\sigma_K^4 R(f'')}{4} h^4 + \frac{R(K)}{nh}, \qquad R(f'') = \int (f'')^2,$$

minimized at

$$h^* = \Big(\frac{R(K)}{\sigma_K^4 R(f'')\, n}\Big)^{1/5} \asymp n^{-1/5}, \qquad \mathrm{MISE}(h^*) \asymp n^{-4/5}. \tag{11.1}$$

(Set the derivative $\sigma_K^4 R(f'')h^3 - R(K)/(nh^2)$ to zero.) The rate $n^{-4/5}$ is slower than the parametric rate n^{-1}: this is the price of not assuming a shape.

Choosing h. The optimal h^* contains the unknown $R(f'')$.

- **Normal reference.** If $f = N(\mu, \sigma^2)$ then $R(f'') = \frac{3}{8\sqrt{\pi}\sigma^5}$, so with the Gaussian kernel $h = 1.06\,\hat\sigma n^{-1/5}$, where $\hat\sigma = \min(s, \mathrm{IQR}/1.34)$.
- **Silverman's rule of thumb:** $h = 0.9\,\hat\sigma n^{-1/5}$, slightly smaller to protect against multimodality.
- **Plug-in (Sheather–Jones):** estimate $R(f'')$ by a pilot kernel estimate of f'' and substitute into (11.1). This is generally the recommended choice; cross-validation is an alternative.

Proof of the normal value of $R(f'')$. For $\sigma = 1$, $\phi''(x) = (x^2-1)\phi(x)$ and $\phi^2 = \frac{1}{2\sqrt{\pi}} \times$ the $N(0, \frac12)$ density. So $R(\phi'') = \frac{1}{2\sqrt{\pi}} E(Z^2-1)^2$ with $Z \sim N(0, \frac12)$, which is $\frac{1}{2\sqrt{\pi}}(\frac34 - 1 + 1) = \frac{3}{8\sqrt{\pi}}$. Scaling gives the factor σ^{-5}. With $R(\phi) = \frac{1}{2\sqrt{\pi}}$ and $\sigma_K = 1$, (11.1) gives $h^* = (4/3)^{1/5}\sigma n^{-1/5} = 1.06\sigma n^{-1/5}$. ■

11.3 Multivariate density estimation

For $\mathbf{X}_i \in \mathbb{R}^p$, $\hat f(\mathbf{x}) = \frac1n \sum_i K_H(\mathbf{x} - \mathbf{X}_i)$ with $K_H(\mathbf{x}) = |H|^{-1/2} K(H^{-1/2}\mathbf{x})$ and a bandwidth matrix H (e.g. $H = \mathrm{diag}(h_1^2, \dots, h_p^2)$, or a plug-in estimate of a full matrix that follows the shape of the data). With a product kernel and equal widths h,

$$\mathrm{MSE}\,\hat f(\mathbf{x}) \approx \Big[\frac{\sigma_K^2 h^2}{2} \sum_j \partial_{jj} f(\mathbf{x})\Big]^2 + \frac{f(\mathbf{x}) R(K)^p}{nh^p},$$

so $h^* \asymp n^{-1/(p+4)}$ and $\mathrm{MSE} \asymp n^{-4/(p+4)}$. This is the **curse of dimensionality**: in $p = 10$ dimensions, a window of side 0.1 in the unit cube contains on average $10^{-10}n$ points. Kernel density estimation is practical in one or two dimensions, occasionally three.

11.4 Confidence intervals and bands

Since bias $\hat f(x)$ and se $\hat f(x)$ are of the same order at h^*, intervals are stated for the smoothed density $f_h(x) = E\hat f(x)$ rather than for $f(x)$.

- **Pointwise, plug-in:** by the CLT, $\sqrt{nh}(\hat f(x) - f_h(x)) \xrightarrow{d} N(0, f(x)R(K))$, so $f_h(x) \in \hat f(x) \pm z_{\alpha/2}\sqrt{\hat f(x) R(K)/(nh)}$.
- **Pointwise, bootstrap:** resample the data, recompute $\hat f^*$ on a grid with the same h, and use $\hat f \pm z_{\alpha/2}\widehat{\mathrm{se}}^*$, the basic interval $(2\hat f - Q^*_{1-\alpha/2}, 2\hat f - Q^*_{\alpha/2})$, or the symmetric interval $\hat f \pm Q_{1-\alpha}(|\hat f^* - \hat f|)$.

- **Simultaneous band:** with $\Delta^* = \max_{x\in\text{grid}}|\hat f^*(x) - \hat f(x)|$ and its $1-\alpha$ quantile q^*, the band $\hat f(x) \pm q^*$ covers f_h on the whole grid with asymptotic probability $1-\alpha$. It is much wider than the pointwise intervals.

Negative lower limits can be truncated at 0 without changing coverage.

The quantity $(\hat f(x) - f_h(x))/\sqrt{\hat f(x)R(K)/(nh)}$ is asymptotically pivotal, which suggests a bootstrap-t interval with $t^* = (\hat f^* - \hat f)/\sqrt{\hat f^* R(K)/(nh)}$. In practice it is unstable: where $\hat f^*(x) \approx 0$ the denominator vanishes, t^* explodes, and the interval becomes absurdly wide. The same problem rules out bands built from studentized maxima. In practice, pointwise intervals on a grid are reported more often than simultaneous bands. Some authors call such a collection a "confidence band", so check which is meant.

11.5 Selected exercises

Motivation. The exercises make the $n^{-4/5}$ rate explicit, show why least squares cross-validation estimates the right criterion, and quantify the claim that the kernel matters much less than the bandwidth.

Exercise 11.5 (The $n^{-4/5}$ rate)

(a) Show that the asymptotic MISE at h^* in (11.1) equals $\frac{5}{4}\{\sigma_K^4 R(f'')\}^{1/5} R(K)^{4/5} n^{-4/5}$.
(b) For x with $f(x) > 0$ and f as in Theorem 11.1, show that $\text{MSE}\,\hat f(x) = O(n^{-4/5})$ for any $h = Cn^{-1/5}$.
(c) Compare with $\text{MSE}(\bar X_n) = \sigma^2/n$.

Solution.

(a) Write $A(h) = ah^4 + c/(nh)$ with $a = \sigma_K^4 R(f'')/4$ and $c = R(K)$. At the minimizer $4ah^{*3} = c/(nh^{*2})$, i.e. $ah^{*4} = c/(4nh^*)$. Hence

$$A(h^*) = \frac{c}{4nh^*} + \frac{c}{nh^*} = \frac{5}{4}\cdot\frac{c}{nh^*}.$$

Substituting $h^* = \{c/(4an)\}^{1/5}$ gives $A(h^*) = \frac{5}{4}(4a)^{1/5}c^{4/5}n^{-4/5}$, and $4a = \sigma_K^4 R(f'')$.

(b) By Theorem 11.1,

$$\text{MSE}\,\hat f(x) = \text{bias}^2 + \text{Var} = \Big(\frac{\sigma_K^2 f''(x)}{2}\Big)^2 h^4 + \frac{f(x)R(K)}{nh} + o\Big(h^4 + \frac{1}{nh}\Big).$$

With $h = Cn^{-1/5}$, $h^4 = C^4 n^{-4/5}$ and $1/(nh) = C^{-1}n^{-4/5}$, so every term is $O(n^{-4/5})$.

(c) $\text{MSE}\,\hat f(x)/\,\text{MSE}(\bar X_n) \asymp n^{1/5} \to \infty$. Halving the MSE requires $2^{5/4} \approx 2.38$ times as many observations for the density at a point, against 2 times for a mean. This is the price of not assuming a shape for f. ■

Exercise 11.7 (Least squares cross-validation)

Let $\hat f_{-i}(x) = \frac{1}{(n-1)h}\sum_{j\neq i} K\big(\frac{x-X_j}{h}\big)$ and

$$\text{CV}(h) = \int \hat f^2 - \frac{2}{n}\sum_{i=1}^n \hat f_{-i}(X_i).$$

Assuming K is bounded and $\int f^2 < \infty$, show that $E\,\mathrm{CV}(h) = \mathrm{MISE}(h) - \int f^2$ for every h. Explain why the leave-one-out estimate is needed.

Solution [Supplement].

Step 1. Expanding the square, $\mathrm{MISE}(h) = E\int \hat f^2 - 2E\int \hat f f + \int f^2$. With $K_h(u) = K(u/h)/h$ and Fubini (all integrands are nonnegative),

$$E\int \hat f(x) f(x)\,dx = \int E\hat f(x)\, f(x)\,dx = \iint K_h(x-t) f(t) f(x)\,dt\,dx.$$

Step 2. For $j \neq i$, X_i and X_j are independent with density f, so $EK_h(X_i - X_j) = \iint K_h(x-t) f(x) f(t)\,dx\,dt$. Hence $E\hat f_{-i}(X_i)$ equals the same double integral, and $E\big[\frac{2}{n}\sum_i \hat f_{-i}(X_i)\big] = 2E\int \hat f f$.

Step 3. Therefore $E\,\mathrm{CV}(h) = E\int \hat f^2 - 2E\int \hat f f = \mathrm{MISE}(h) - \int f^2$. The last term does not depend on h, so minimizing CV targets an unbiased estimate of MISE up to a constant.

Using $\hat f(X_i) = \frac{K(0)}{nh} + \frac{n-1}{n}\hat f_{-i}(X_i)$ instead would subtract the extra term $\frac{2K(0)}{nh}$, coming from $j = i$, which tends to ∞ as $h \to 0$. The criterion would then be driven to $h \to 0$: every point would be "predicted" by its own spike. ■

Exercise 11.8 (How much does the kernel matter?)

(a) Show that $\sigma_K R(K)$ is unchanged by rescaling $K_c(u) = K(u/c)/c$, and that the optimal AMISE is proportional to $\{\sigma_K R(K)\}^{4/5}$.
(b) Compute $\sigma_K R(K)$ for the Gaussian, Epanechnikov and uniform kernels, and the relative sample sizes needed for equal optimal AMISE.
(c) Derive the normal-reference bandwidth for the Epanechnikov kernel.

Solution [Supplement].

(a) Substituting $u = cv$ gives $\sigma^2_{K_c} = \int u^2 K(u/c)/c\,du = c^2\sigma_K^2$ and $R(K_c) = \int K(u/c)^2/c^2\,du = R(K)/c$. So $\sigma_{K_c} R(K_c) = \sigma_K R(K)$. A rescaled kernel only changes the meaning of h. By Exercise 11.5(a), the optimal AMISE is

$$\frac{5}{4} R(f'')^{1/5} (\sigma_K^4)^{1/5} R(K)^{4/5} n^{-4/5} = \frac{5}{4} R(f'')^{1/5} \{\sigma_K R(K)\}^{4/5} n^{-4/5}.$$

(b) Gaussian: $\sigma_K = 1$, $R(K) = \frac{1}{2\sqrt{\pi}}$, product 0.2821. Epanechnikov $\frac{3}{4}(1-u^2)_+$: $\sigma_K^2 = \frac{3}{4}\int_{-1}^1 (u^2 - u^4)\,du = \frac{1}{5}$ and $R(K) = \frac{9}{16}\int_{-1}^1 (1-u^2)^2\,du = \frac{3}{5}$, product $\frac{3}{5\sqrt{5}} = 0.2683$. Uniform: $\sigma_K^2 = \frac{1}{3}$, $R(K) = \frac{1}{2}$, product 0.2887. Equal AMISE means $\{\sigma_K R(K)\}^{4/5} n^{-4/5}$ is equal, so the required sample sizes are proportional to $\sigma_K R(K)$. Relative to the Epanechnikov kernel, the Gaussian needs $0.2821/0.2683 = 1.05$ times as many observations and the uniform 1.08 times. The choice of K costs at most a few percent; the choice of h can cost far more.

(c) With $R(f'') = \frac{3}{8\sqrt{\pi}\sigma^5}$ (Section 11.2), (11.1) gives

$$h^* = \Big(\frac{3/5}{(1/25)\cdot 3/(8\sqrt{\pi}\sigma^5)\cdot n}\Big)^{1/5} = (40\sqrt{\pi})^{1/5}\sigma n^{-1/5} = 2.34\,\sigma n^{-1/5}.$$

The Epanechnikov bandwidth is larger than the Gaussian $1.06\,\sigma n^{-1/5}$ only because the two kernels have different scales ($\sigma_K = 0.447$ versus 1), as (a) explains. ■